Generalized Additive Model for Council District VII (2023 - 2025 Sales)

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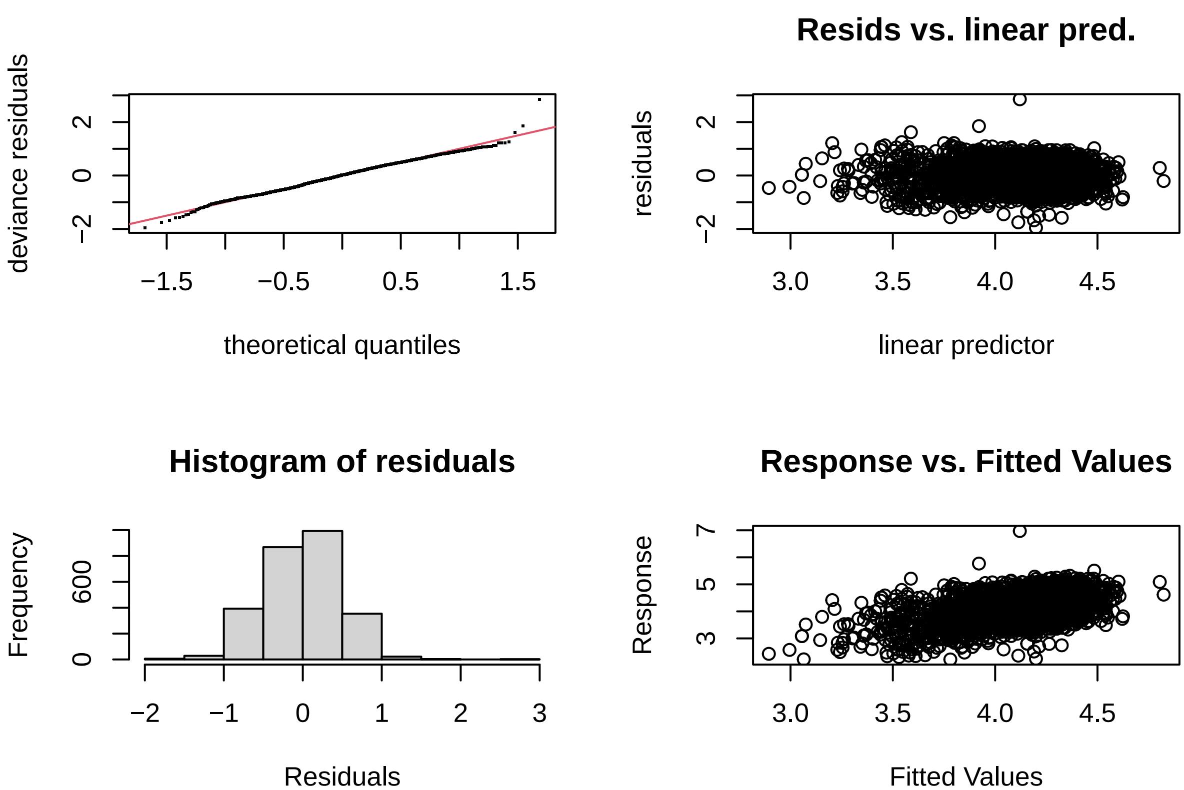
Contents

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          4.049470   0.038372 105.532  < 2e-16 ***
## arterial            -0.011638   0.035959  -0.324 0.746228
## single_family1story  0.065169   0.033979   1.918 0.055233 .
## half_duplex1story    0.016341   0.122719   0.133 0.894080
## half_duplex2story    0.153120   0.148419   1.032 0.302319
## smallest             0.168037   0.033056   5.083 3.97e-07 ***
## small                0.086219   0.024505   3.518 0.000442 ***
## large               -0.083806   0.035718  -2.346 0.019035 *
## largest             -0.099270   0.079107  -1.255 0.209634
## two_plus_baths       0.105653   0.102191   1.034 0.301290
## powder_room          0.006709   0.056489   0.119 0.905474
## good                 0.129101   0.095057   1.358 0.174534
```

```
## fair                -0.160447   0.022853  -7.021 2.80e-12 ***
## poor                -0.283098   0.053614  -5.280 1.40e-07 ***
## very_poor_unsound   -0.110187   0.124978  -0.882 0.378047
## qabove_avg           0.271934   0.133729   2.033 0.042105 *
## qbelow_avg          -0.055761   0.023747  -2.348 0.018941 *
## crawl                0.206426   0.076875   2.685 0.007294 **
## slab                -0.153253   0.084356  -1.817 0.069369 .
## fireplace            0.045869   0.025607   1.791 0.073367 .
## hot_water            0.124984   0.050723   2.464 0.013802 *
## elec_baseboard      -0.501880   0.289324  -1.735 0.082917 .
## wall                -0.192728   0.102977  -1.872 0.061376 .
## central_air          0.140661   0.026372   5.334 1.04e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                        edf Ref.df      F p-value
## s(sresb_yearbuilt) 7.256      9 22.272  <2e-16 ***
## s(ln_garagespaces) 1.655      3 17.327  <2e-16 ***
## s(months)          2.304      9  6.984  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.213   Deviance explained = 22.3%
## -REML = 1851.4  Scale est. = 0.22483   n = 2667
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-0.0008041874,-8.993674e-07]
```

```
## (score 1851.429 & scale 0.2248328).
## Hessian positive definite, eigenvalue range [0.8540881,1321.512].
## Model rank =  45 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                      k'  edf k-index p-value
## s(sresb_yearbuilt) 9.00 7.26    0.86  <2e-16 ***
## s(ln_garagespaces) 3.00 1.65    0.88  <2e-16 ***
## s(months)          9.00 2.30    0.86  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod   prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl> <dbl>
## 1  2667     0.000882    0.00102    0.000788  1.29  46.5 -1.02
```

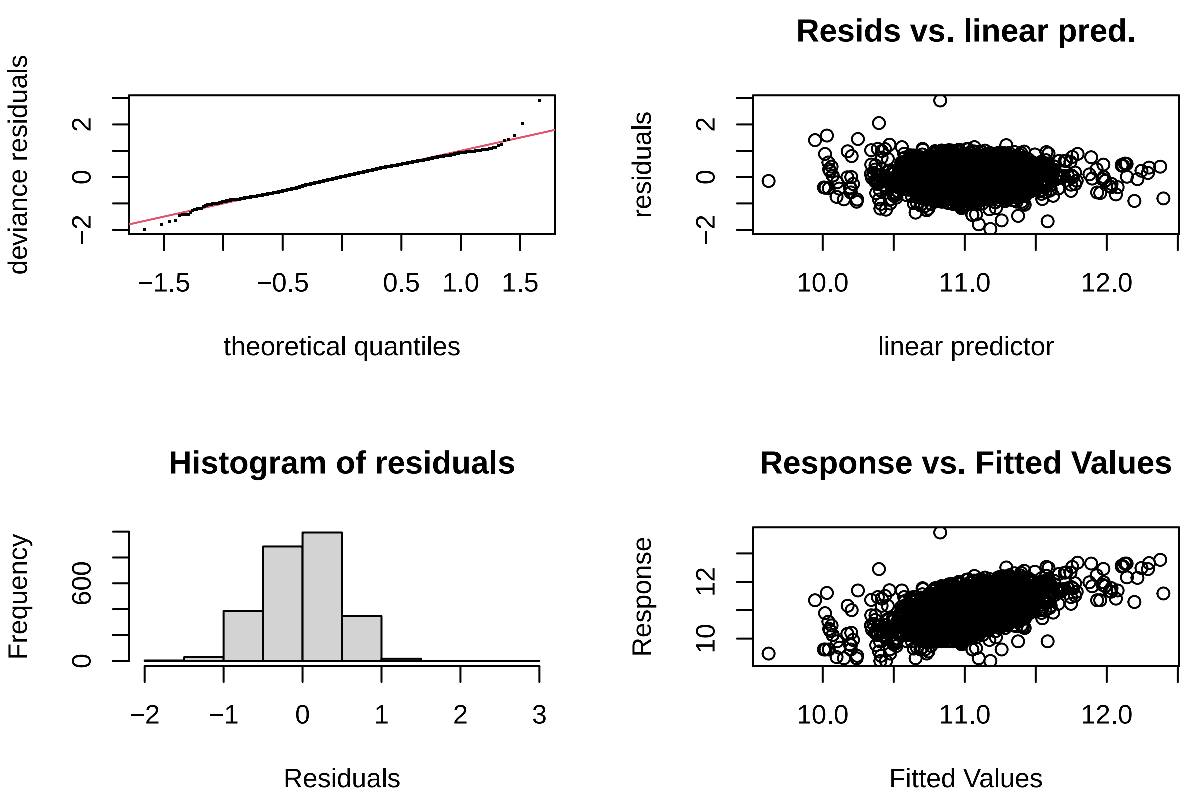
Table 1: District 7 LN_SPPSF NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2667	0.0009588	0.0010997	0.0008516	1.291409	45.82434	-0.0009519

Model II

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.054520   0.041217 268.202  < 2e-16 ***
## arterial            -0.003429   0.035696  -0.096  0.92349
## single_family1story  0.048708   0.033748   1.443  0.14906
## half_duplex1story   -0.125232   0.124105  -1.009  0.31303
## half_duplex2story    0.016541   0.148310   0.112  0.91121
## smallest            -0.015127   0.071344  -0.212  0.83210
## small               -0.001202   0.036254  -0.033  0.97355
## large                0.008447   0.048780   0.173  0.86253
## largest              0.050503   0.096933   0.521  0.60240
```

```
## two_plus_baths       0.262406   0.133129   1.971  0.04882 *
## powder_room          0.054846   0.061367   0.894  0.37154
## good                 0.129921   0.093850   1.384  0.16637
## fair                -0.159079   0.022552  -7.054 2.21e-12 ***
## poor                -0.278177   0.052878  -5.261 1.55e-07 ***
## very_poor_unsound   -0.114630   0.123550  -0.928  0.35360
## qabove_avg           0.295377   0.132931   2.222  0.02637 *
## qbelow_avg          -0.070731   0.023573  -3.001  0.00272 **
## crawl                0.167270   0.075834   2.206  0.02749 *
## slab                -0.172936   0.083740  -2.065  0.03901 *
## fireplace            0.053516   0.025283   2.117  0.03438 *
## hot_water            0.119235   0.050044   2.383  0.01726 *
## elec_baseboard      -0.533451   0.291562  -1.830  0.06742 .
## wall                -0.220702   0.101656  -2.171  0.03002 *
## central_air          0.136215   0.026033   5.232 1.81e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                             edf Ref.df      F  p-value
## s(ln_landratio)           3.997      9  8.699  < 2e-16 ***
## s(ln_base_bldgsize_ratio) 1.513      9  1.426 0.000283 ***
## s(sresb_yearbuilt)        6.134      9 13.056  < 2e-16 ***
## s(ln_garagespaces)        1.573      3 12.272  < 2e-16 ***
## s(months)                 2.430      9  7.930  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =   0.27   Deviance explained = 28.1%
## -REML =   1818  Scale est. = 0.21844   n = 2667
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 5 iterations.
## Gradient range [-0.001733453,1.014086e-05]
## (score 1817.979 & scale 0.2184392).
## Hessian positive definite, eigenvalue range [0.4914681,1321.514].
## Model rank =  63 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'  edf k-index p-value
## s(ln_landratio)            9.00 4.00    0.93  <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.00 1.51    0.96    0.02 *
## s(sresb_yearbuilt)        9.00 6.13    0.88  <2e-16 ***
## s(ln_garagespaces)        3.00 1.57    0.88  <2e-16 ***
## s(months)                 9.00 2.43    0.87  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2667        0.908       1.03       0.833  1.24  41.7 -0.913
```

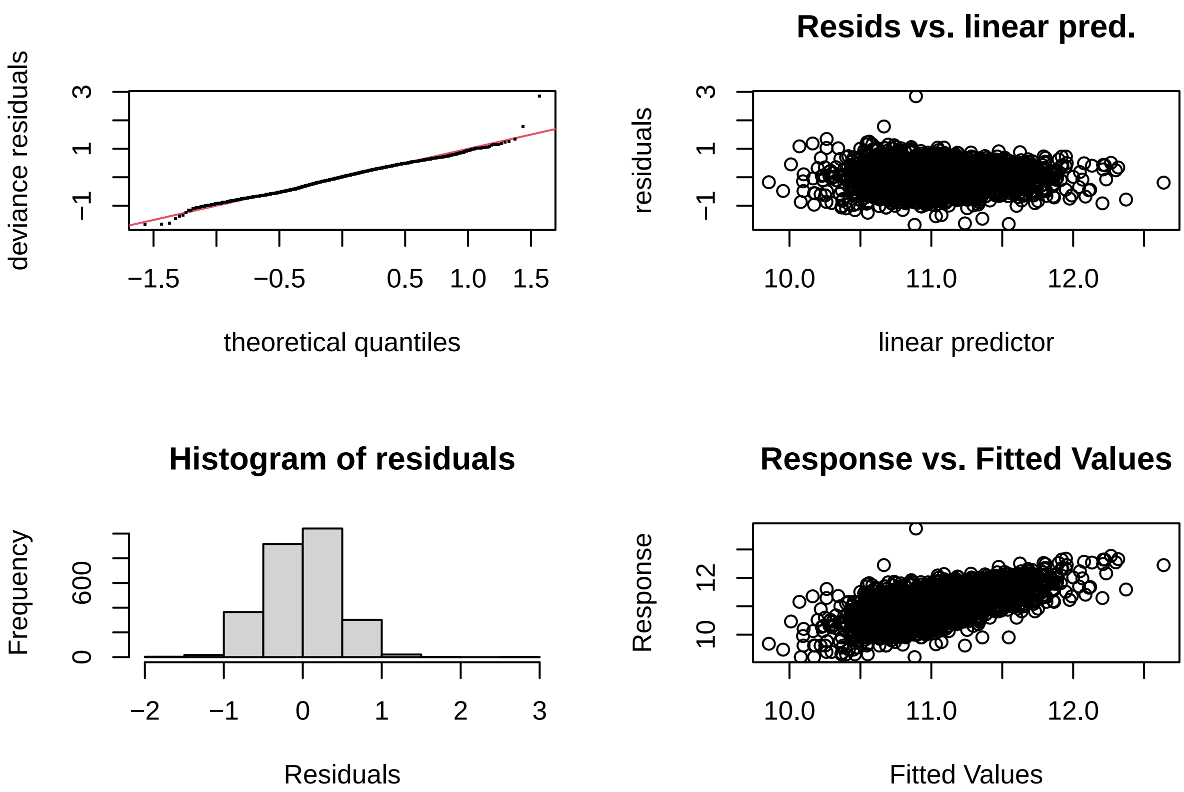
Table 2: District 7 LN_PRICE (No ECFs) NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2667	0.9848801	1.114926	0.8999719	1.238846	41.05806	-0.833201

Model III

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.571965   0.066826 173.166  < 2e-16 ***
## arterial            -0.010290   0.034707  -0.296 0.766877
## single_family1story  0.033980   0.032087   1.059 0.289690
```

```
## half_duplex1story   -0.138423   0.118925  -1.164 0.244552
## half_duplex2story    0.030200   0.142754   0.212 0.832475
## smallest             0.002273   0.073002   0.031 0.975160
## small                0.004765   0.037475   0.127 0.898834
## large                0.034515   0.052013   0.664 0.507017
## largest              0.049022   0.102814   0.477 0.633543
## two_plus_baths       0.239509   0.153760   1.558 0.119432
## powder_room          0.095296   0.062502   1.525 0.127457
## good                 0.100533   0.089269   1.126 0.260190
## fair                -0.140540   0.021554  -6.520 8.39e-11 ***
## poor                -0.269916   0.050320  -5.364 8.86e-08 ***
## very_poor_unsound   -0.067868   0.117008  -0.580 0.561943
## qabove_avg           0.128159   0.135412   0.946 0.344012
## qbelow_avg          -0.080985   0.023344  -3.469 0.000530 ***
## crawl                0.153401   0.072416   2.118 0.034243 *
## slab                -0.116755   0.079486  -1.469 0.141985
## fireplace            0.078280   0.024815   3.155 0.001626 **
## hot_water            0.065866   0.047916   1.375 0.169364
## elec_baseboard      -0.402704   0.273441  -1.473 0.140945
## wall                -0.226962   0.096928  -2.342 0.019278 *
## central_air          0.094424   0.024800   3.807 0.000144 ***
## ecf7R702            -0.103164   0.066049  -1.562 0.118424
## ecf7R703            -0.677201   0.081823  -8.276  < 2e-16 ***
## ecf7R704            -0.495857   0.057954  -8.556  < 2e-16 ***
## ecf7R705            -0.499398   0.054351  -9.188  < 2e-16 ***
## ecf7R706            -0.481227   0.089703  -5.365 8.82e-08 ***
## ecf7R707            -0.545899   0.095918  -5.691 1.40e-08 ***
## ecf7R708            -0.587695   0.066841  -8.792  < 2e-16 ***
## ecf7R709            -0.563766   0.065001  -8.673  < 2e-16 ***
## ecf7R710            -0.352006   0.067216  -5.237 1.76e-07 ***
## ecf7R711            -0.632347   0.066059  -9.572  < 2e-16 ***
## ecf7R712            -0.491532   0.083022  -5.921 3.63e-09 ***
## ecf7R713            -0.540426   0.076230  -7.089 1.73e-12 ***
## ecf7R714            -0.746348   0.207277  -3.601 0.000323 ***
## ecf7R715            -0.755425   0.109309  -6.911 6.03e-12 ***
## ecf7R716            -1.027718   0.081033 -12.683  < 2e-16 ***
## ecf7R717            -0.887685   0.082539 -10.755  < 2e-16 ***
## ecf7R718            -0.779903   0.067971 -11.474  < 2e-16 ***
## ecf7R719            -0.422117   0.059577  -7.085 1.78e-12 ***
## ecf7R720            -0.884456   0.142283  -6.216 5.91e-10 ***
## ecf7R721            -0.223498   0.161859  -1.381 0.167452
## ecf7R722            -0.624882   0.117217  -5.331 1.06e-07 ***
## ecf7R723            -0.172996   0.094008  -1.840 0.065847 .
## ecf7R724            -0.729947   0.095852  -7.615 3.65e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                             edf Ref.df     F  p-value
## s(ln_landratio)           1.638      9 1.221 0.001076 **
## s(ln_base_bldgsize_ratio) 2.580      9 1.771 0.000191 ***
## s(sresb_yearbuilt)        5.031      9 4.348  < 2e-16 ***
## s(ln_garagespaces)        1.455      3 8.464 8.66e-07 ***
## s(months)                 2.329      9 9.449  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =   0.35   Deviance explained = 36.4%
## -REML = 1689.5  Scale est. = 0.19456   n = 2667
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-7.136982e-07,3.65512e-06]
## (score 1689.539 & scale 0.1945601).
## Hessian positive definite, eigenvalue range [0.2160219,1310.008].
## Model rank =  86 / 86
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                             k'  edf k-index p-value
## s(ln_landratio)           9.00 1.64    0.99    0.38
## s(ln_base_bldgsize_ratio) 9.00 2.58    0.98    0.16
## s(sresb_yearbuilt)        9.00 5.03    0.95  <2e-16 ***
## s(ln_garagespaces)        3.00 1.46    0.96    0.01 **
## s(months)                 9.00 2.33    0.93  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

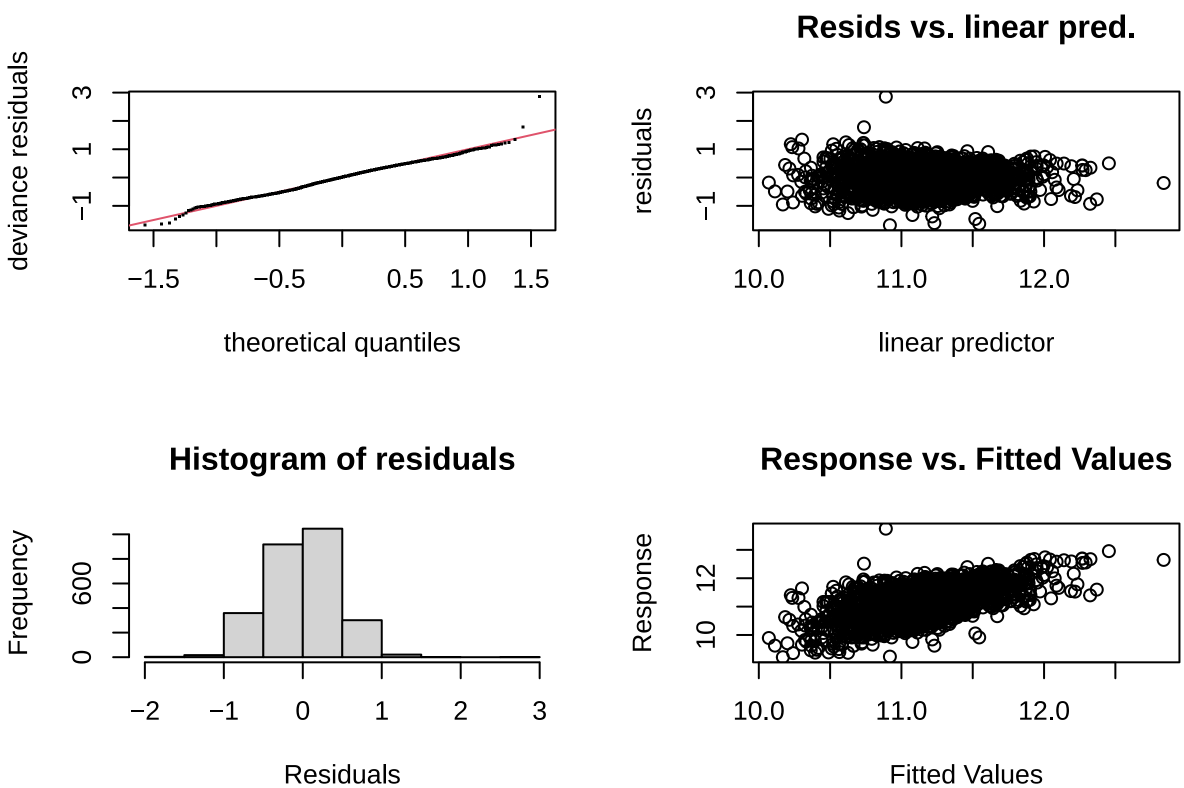
```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2667        0.908       1.02       0.845  1.20  39.2 -0.715
```

Model IV

```
##
```

```
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.651389   0.066627 174.875  < 2e-16 ***
## arterial            -0.009465   0.034679  -0.273 0.784925
## single_family1story  0.032301   0.030778   1.049 0.294059
## half_duplex1story   -0.137955   0.118226  -1.167 0.243365
## half_duplex2story    0.027643   0.142711   0.194 0.846426
## smallest             0.004712   0.073036   0.065 0.948560
## small                0.005078   0.037503   0.135 0.892294
## large                0.034409   0.052095   0.660 0.508997
## largest              0.047810   0.102905   0.465 0.642254
## two_plus_baths       0.242768   0.154195   1.574 0.115511
## powder_room          0.096772   0.062558   1.547 0.122005
## good                 0.096363   0.089182   1.081 0.280011
## fair                -0.140914   0.021514  -6.550 6.91e-11 ***
## poor                -0.269358   0.050292  -5.356 9.26e-08 ***
## very_poor_unsound   -0.062846   0.116850  -0.538 0.590736
## qabove_avg           0.124615   0.135360   0.921 0.357335
## qbelow_avg          -0.080649   0.023332  -3.457 0.000556 ***
## crawl                0.150523   0.072328   2.081 0.037520 *
## slab                -0.115293   0.079419  -1.452 0.146707
## fireplace            0.077747   0.024788   3.136 0.001729 **
## hot_water            0.065498   0.047899   1.367 0.171606
## elec_baseboard      -0.403372   0.273356  -1.476 0.140165
## wall                -0.229589   0.096864  -2.370 0.017850 *
## central_air          0.094259   0.024788   3.803 0.000146 ***
## ecf7R702            -0.103366   0.066016  -1.566 0.117523
## ecf7R703            -0.673594   0.081666  -8.248 2.52e-16 ***
## ecf7R704            -0.493652   0.057867  -8.531  < 2e-16 ***
## ecf7R705            -0.498856   0.054313  -9.185  < 2e-16 ***
## ecf7R706            -0.479759   0.089621  -5.353 9.39e-08 ***
## ecf7R707            -0.543100   0.095853  -5.666 1.62e-08 ***
## ecf7R708            -0.585746   0.066740  -8.776  < 2e-16 ***
## ecf7R709            -0.561051   0.064890  -8.646  < 2e-16 ***
## ecf7R710            -0.351157   0.067089  -5.234 1.79e-07 ***
## ecf7R711            -0.630529   0.066029  -9.549  < 2e-16 ***
## ecf7R712            -0.489770   0.082936  -5.905 3.97e-09 ***
## ecf7R713            -0.537330   0.076192  -7.052 2.25e-12 ***
## ecf7R714            -0.737407   0.207099  -3.561 0.000377 ***
## ecf7R715            -0.749816   0.109208  -6.866 8.23e-12 ***
## ecf7R716            -1.023895   0.080968 -12.646  < 2e-16 ***
## ecf7R717            -0.883386   0.082474 -10.711  < 2e-16 ***
## ecf7R718            -0.775429   0.067872 -11.425  < 2e-16 ***
## ecf7R719            -0.420244   0.059518  -7.061 2.11e-12 ***
## ecf7R720            -0.884076   0.142252  -6.215 5.96e-10 ***
## ecf7R721            -0.218821   0.161802  -1.352 0.176363
```

```
## ecf7R722            -0.621684   0.117181  -5.305 1.22e-07 ***
## ecf7R723            -0.169203   0.093951  -1.801 0.071823 .
## ecf7R724            -0.725253   0.095821  -7.569 5.19e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                               edf Ref.df     F  p-value
## s(ln_landratio)            1.6521      9 1.219 0.001115 **
## s(ln_base_bldgsize_ratio) 2.6064      9 1.791 0.000179 ***
## s(sresb_yearbuilt)         5.0638      9 4.396  < 2e-16 ***
## s(ln_garagespaces)         1.4614      3 8.670 5.25e-07 ***
## s(months)                  0.1709      9 0.022 0.275284
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.337   Deviance explained = 35.1%
## -REML = 1686.1  Scale est. = 0.19448   n = 2667
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-6.137629e-07,1.967995e-06]
## (score 1686.069 & scale 0.194475).
## Hessian positive definite, eigenvalue range [0.01370283,1310.007].
## Model rank =  86 / 86
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
```

```
##                                k'   edf k-index p-value
## s(ln_landratio)           9.000 1.652    0.99   0.290
## s(ln_base_bldgsize_ratio) 9.000 2.606    0.98   0.090 .
## s(sresb_yearbuilt)        9.000 5.064    0.95   0.005 **
## s(ln_garagespaces)        3.000 1.461    0.96   0.005 **
## s(months)                 9.000 0.171    0.93  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2667        0.986       1.10       0.913  1.20  38.4 -0.729
```